THE GABRIEL-POPESCU ADJUNCTION IS AN EQUIVALENCE ON THE MODULE SIDE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The adjunction of the two functors. Same chapter, the construction of the functors F and G and their adjunction established earlier. Supplies the pair of functors whose derived functors are compared and the identity of the composite on the module side.
- Fully faithful adjunctions. Categories, Lemma categories-lemma-adjoint-fully-faithful. Supplies that an adjunction whose unit or counit is an isomorphism is fully faithful, which is how full faithfulness of the derived functor is concluded.
- K-injective representatives. Same chapter, Theorem theorem-K-injective-embedding-grothendieck. Supplies that the derived functor can be computed on a K-injective representative, which is what makes the composite computation available.

Source reference. Original theorem 0F5U: sources/stacks-injectives/pinned-original-source.tex, source lines 2878–2878. Navigation pointer invoked at proof line 2928; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:injectives.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original theorem 079P: sources/stacks-injectives/pinned-original-source.tex, source lines 2024–2024. Navigation pointer invoked at proof line 2942; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 070Y: sources/stacks-derived/pinned-original-source.tex, source lines 9964–9964. Navigation pointer invoked at proof line 2944; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 015F: sources/stacks-derived/pinned-original-source.tex, source lines 6101–6101. Navigation pointer invoked at proof line 2945; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09T5: sources/stacks-derived/pinned-original-source.tex, source lines 9813–9813. Navigation pointer invoked at proof line 2946; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07RB: sources/stacks-categories/pinned-original-source.tex, source lines 3657–3657. Navigation pointer invoked at proof line 2893; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be a Grothendieck abelian category. Let R, F, G be as in the Gabriel-Popescu theorem (Theorem 0F5U). Then we obtain derived functors*

$$RG : D(\mathcal{A}) \to D(Mod_R) \quad \text{and} \quad F : D(Mod_R) \to D(\mathcal{A})$$

such that F is left adjoint to RG, RG is fully faithful, and $F \circ RG = id$.

Proof. The existence and adjointness of the functors follows from Theorems 0F5U and 079P and Derived Categories, Lemmas 070Y, 015F, and 09T5. The statement $F \circ RG = \mathrm{id}$ follows because we can compute RG on an object of $D(\mathcal{A})$ by applying G to a suitable representative complex $I^\bullet$ (for example a K-injective one) and then $F(G(I^\bullet)) = I^\bullet$ because $F \circ G = \mathrm{id}$. Fully faithfulness of RG follows from this by Categories, Lemma 07RB. □